

# INFLUENCE OF SURFACE LAYER ON CRYSTAL ORDERING NEAR THE MELTING POINT

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Inhomogeneous crystal ordering in a semi-infinite (flat surface) crystals is investigated within the framework of the crystalline order theory in which the tensor local order parameter (cubic nonor) is used. The effects due to the surface are treated by introducing the external field  $h_{\alpha\beta\gamma\delta}^e$ , which depends on the deformation of the crystal lattice near crystal surface. The field  $h_{\alpha\beta\gamma\delta}^e$  acts on the surface only. As the result the connection between  $h_{\alpha\beta\gamma\delta}^e$  and a phenomenological constant  $C$ , introduced by the authors and co-worker previously (*Zh. Eksp. Teor. Fiz.* **81**, 2198 (1981)) is obtained.

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As shown in [1], a structure of the crystal surface considerably influences the properties of the crystal near its melting point  $T_m$ . In the first part of paper [1] a model neglecting deformation in the crystal surface had been considered. It was shown, that in a small vicinity of the melting temperature  $T_m$  a part of the crystal near its surface would be practically in a liquid state, whereas the bulk would preserve the crystalline structure. In this case, between melted and unmelted parts of the crystal, there would be relatively sharp boundary wall, similar to the domain wall in the theory of magnetism. The width of the melted part of the crystal would be proportional to  $|\ln|\tau||$ , where  $\tau = (T - T_m)/T_m$ , i.e. it would increase to  $+\infty$ , when  $T \rightarrow T_m$ . As the result, the crystal without any deformation in the surface layer becomes liquid at  $T = T_m$ . Therefore the metastable state of the overheated crystal does not appear in this case.

On the other hand, the overheated crystals were observed in experiments [2, 3]. The rate of the overheating of real crystals is of order of  $10^{-3}$  [3].

It is well known that in real crystals the structure of the surface layer is slightly different from the bulk structure. It is due to e.g. an asymmetry of forces acting in the perpendicular direction to the surface plane, on atoms forming the surface layer. For the temperatures

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far below  $T_m$ , the change of structure of the surface layer does not influence distinctly the crystal ordering in bulk. A big difference between thermodynamical potentials of crystal and liquid phases exists, and a small perturbation connected with the existence of different structure in the thin surface layer does not change the bulk properties.

The situation changes drastically in a close vicinity of the melting point  $T_m$ , where the difference of the thermodynamical potentials between the liquid and the crystal  $\Delta\phi$  is small. In this case the small change of the structure of the surface layer considerably influences the crystal order far from the surface. In other words a small change of condition of the crystal surface has an important influence on the ordering in the whole crystal only when  $T$  approaches  $T_m$ .

The difference between the structure of the crystal layer and the bulk was taken into account in a phenomenological way in the second part of paper [1]. The crystalline order near melting point  $T_m$  depends on a value of a phenomenological constant  $C$  introduced in [1]. Namely, the width of the melted part of the crystal was found to be proportional to  $|\ln|C||$  for  $|\tau| \ll C^2$ ,  $C \neq 0$ , and to have finite value for  $\tau = 0$ , i.e. for  $T = T_m$ .

The aim of this paper was to find the physical meaning of the phenomenological constant  $C$  introduced in [1], i.e. its connection with the deformation of the crystal surface layer.

In the following discussion we will consider only cubic crystals. The order parameter of the local crystal ordering is the fourth-rank irreducible tensor (cubic nonor)  $\tilde{K}_{\alpha\beta\gamma\delta}$  [1, 4, 5]. Here the tilde means that the tensor  $\tilde{K}_{\alpha\beta\gamma\delta}$  is irreducible (anisotropic). The effective hamiltonian of the crystalline order was chosen in the form

$$H\{\tilde{K}_{\alpha\beta\gamma\delta}(\bar{r})\} = - \int \tilde{I}(|\bar{r} - \bar{r}'|) \tilde{K}_{\alpha\beta\gamma\delta}(\bar{r}) \tilde{K}_{\alpha\beta\gamma\delta}(\bar{r}') d\bar{r} d\bar{r}'. \quad (1)$$

In the mean field approximation (MFA) the effective hamiltonian describing the interaction of the cluster located in point  $\bar{r}$  with the mean field  $\tilde{h}_{\alpha\beta\gamma\delta}(\bar{r}) \equiv \tilde{h}_{\alpha\beta\gamma\delta}(x, y, z)$  takes the form:

$$H_{\text{MFA}}(\hat{g}; \bar{r}) = - \tilde{K}_{\alpha\beta\gamma\delta}(\hat{g}) \tilde{h}_{\alpha\beta\gamma\delta}(\bar{r}), \quad (2)$$

where  $\hat{g}$  is a rotation in three-dimensional space [4, 5]. The field  $\tilde{h}_{\alpha\beta\gamma\delta}(\bar{r})$  is due to the interactions of all clusters with the one located at  $\bar{r}$  and has a cubic symmetry as well [1]:

$$\tilde{h}_{\alpha\beta\gamma\delta}(\bar{r}) = \tilde{h}_{\alpha\beta\gamma\delta}^c(\bar{r}) = \int d\bar{r}' \tilde{I}(|\bar{r} - \bar{r}'|) \langle \tilde{K}_{\alpha\beta\gamma\delta}(\bar{r}') \rangle. \quad (3)$$

Index  $c$  in  $\tilde{h}_{\alpha\beta\gamma\delta}^c$  denotes the cubic symmetry of this field. In the above formula the integration is carried out over the whole volume of the crystal ( $x \geq 0$ ,  $-\infty < y < +\infty$ ,  $-\infty < z < +\infty$ ); the plane  $x = 0$  is the crystal surface.

Let us consider the case when the crystal order depends only on a coordinate  $x$  perpendicular to the surface. It means that the average  $\langle \tilde{K}_{\alpha\beta\gamma\delta}(\bar{r}) \rangle$  does not depend on coordinates  $y$  and  $z$ <sup>1</sup>. Then instead of (3) we have:

$$\tilde{h}_{\alpha\beta\gamma\delta}(x) = \int_0^{+\infty} I(|x - x'|) \langle \tilde{K}_{\alpha\beta\gamma\delta}(x') \rangle dx', \quad (4)$$

<sup>1</sup> Simultaneously an orientation  $\hat{g}$  of tensors  $\tilde{K}_{\alpha\beta\gamma\delta}(\hat{g}; \bar{r})$  may be different for various points laying on the same plane parallel to the surface.



where

$$I(|x-x'|) \equiv \int_{-\infty}^{+\infty} dy \int_{-\infty}^{+\infty} dz \hat{l}(|\vec{r}-\vec{r}'|).$$

The statistical-mechanic average  $\langle \tilde{K}_{\alpha\beta\gamma\delta} \rangle$  of  $\tilde{K}_{\alpha\beta\gamma\delta}$  is taken over rotations  $\hat{g}$ :

$$\begin{aligned} \langle \tilde{K}_{\alpha\beta\gamma\delta} \rangle &= Z^{-1} \int d\hat{g} \tilde{K}_{\alpha\beta\gamma\delta}(\hat{g}) \exp \{ -H_{\text{MFA}}(\hat{g})/T \}, \\ Z &= \int d\hat{g} \exp \{ -H_{\text{MFA}}(\hat{g})/T \}. \end{aligned} \quad (5)$$

Near the crystal surface the elastic deformation of the cluster described by the strain tensor  $u_{\alpha\beta}$  contributes to the total energy according to the Hook's law [6]:

$$H^e(\hat{g}) = -\Lambda_{\alpha\beta\gamma\delta}(\hat{g}) u_{\alpha\beta} u_{\gamma\delta}, \quad (6)$$

where  $\Lambda_{\alpha\beta\gamma\delta}$  is the tensor of elastic constants.

The deformation is due to forces acting in the perpendicular direction to the surface. The forces result from interactions between atoms forming the surface layer and those laying in the volume of the crystal. They are identical in the whole of the surface plane. As a result the vector  $\vec{s}$  describing shifts of the atoms laying on the surface is the same on the whole surface plane, and has only one component  $s_x$ . Thus the strain tensor  $u_{\alpha\beta}$  has the form:

$$u_{\alpha\beta} = \begin{cases} 0 & \text{for } \alpha \neq x \text{ or } \beta \neq x \\ u_{xx} \neq 0 & \text{for } \alpha = x \text{ and } \beta = x. \end{cases} \quad (7)$$

Thus instead of (6) one may write:

$$H^e(\hat{g}) = -\Lambda_{xxxx}(\hat{g}) u_{xx} u_{xx}. \quad (8)$$

The irreducible part  $\tilde{\Lambda}_{\alpha\beta\gamma\delta}$  of tensor  $\Lambda_{\alpha\beta\gamma\delta}$  is related in a simple way to our order parameter  $\tilde{K}_{\alpha\beta\gamma\delta}$

$$\tilde{\Lambda}_{\alpha\beta\gamma\delta}(\hat{g}) = \varepsilon \cdot \tilde{K}_{\alpha\beta\gamma\delta}(\hat{g}), \quad (9)$$

where  $\varepsilon$  — computational constant.

The reducible part of the tensor  $\Lambda_{\alpha\beta\gamma\delta}$  is an isotropic tensor, and does not depend on  $\hat{g}$ . It gives rise only to constant in hamiltonian (8). From formulas (2), (6), (9) one obtains

$$H_{\text{MFA}}(\hat{g}) = -\tilde{K}_{\alpha\beta\gamma\delta}(\hat{g}) h_{\alpha\beta\gamma\delta} = -\tilde{K}_{\alpha\beta\gamma\delta}(\hat{g}) h_{\alpha\beta\gamma\delta}^e - \tilde{K}_{\alpha\beta\gamma\delta}(\hat{g}) h_{\alpha\beta\gamma\delta}^0, \quad (10)$$

where

$$h_{\alpha\beta\gamma\delta}^e = \varepsilon \cdot u_{\alpha\beta} u_{\gamma\delta}. \quad (11)$$

Here from (8)

$$h_{\alpha\beta\gamma\delta}^e = \begin{cases} h_{xxxx}^e = \varepsilon \cdot u_{xx} u_{xx} & \text{for } \alpha = \beta = \gamma = \delta = x \\ 0 & \text{otherwise.} \end{cases} \quad (12)$$

The fields  $\tilde{h}_{\alpha\beta\gamma\delta}^c$  (see (4)) and  $h_{\alpha\beta\gamma\delta}^c$  (see (7), (11)) have different symmetries. Moreover  $\tilde{h}_{\alpha\beta\gamma\delta}^c$  is an irreducible tensor whereas  $h_{\alpha\beta\gamma\delta}^c$  is reducible. Subsequently we intend to obtain scalar equations and therefore we have to separate the irreducible, cubic-symmetry part  $\tilde{h}_{\alpha\beta\gamma\delta}^{c(c)}$  from the tensor  $h_{\alpha\beta\gamma\delta}^c$ . For this purpose a symmetrization operator  $\hat{R}$  over the cubic-symmetry group ([7]) has been introduced.

$$\hat{R} \equiv \frac{1}{N(G_k)} \sum_{\hat{g}_k \in G_k} \hat{g}_k, \quad (13)$$

where  $\hat{g}_k$  denotes the element of the cubic-symmetry group  $G_k$ , and  $N(G_k) = 48$  for  $G_k \equiv m3m$ . Then

$$h_{\alpha\beta\gamma\delta}^{c(c)} = \hat{R} h_{\alpha\beta\gamma\delta}^c. \quad (14)$$

The field  $h_{\alpha\beta\gamma\delta}^c$  are due to deformations at the surface and we assume it is constant. Further on we will consider the irreducible, cubic-symmetry field  $\tilde{h}_{\alpha\beta\gamma\delta}$  which is the sum of the internal field  $\tilde{h}_{\alpha\beta\gamma\delta}^c$  due to neighbouring clusters and the external one  $\tilde{h}_{\alpha\beta\gamma\delta}^{c(c)}$  associated with the elastic deformation.

$$h_{\alpha\beta\gamma\delta}(x) = \tilde{h}_{\alpha\beta\gamma\delta}^c(x) + \tilde{h}_{\alpha\beta\gamma\delta}^{c(c)}(x). \quad (15)$$

The field  $\tilde{h}_{\alpha\beta\gamma\delta}^{c(c)}$  acts only near the crystal surface.

When the crystal surface is a crystallographic plane, e.g. (001), individual fields appearing in (15) are proportional to the same standard tensor  $\tilde{\Pi}_{\alpha\beta\gamma\delta}$  (cubic nonor).  $\tilde{\Pi}_{\alpha\beta\gamma\delta}$  is the irreducible part of  $\Pi_{\alpha\beta\gamma\delta}$ , where

$$\Pi_{\alpha\beta\gamma\delta} = \sum_{i=1}^3 n_{\alpha}^{(i)} n_{\beta}^{(i)} n_{\gamma}^{(i)} n_{\delta}^{(i)}, \quad (16)$$

with  $\bar{n}^{(1)}$ ,  $\bar{n}^{(2)}$ ,  $\bar{n}^{(3)}$  — three orthonormal vectors. Then the formula (15) can be rewritten using the scalar amplitudes

$$\begin{aligned} h(x) &= h^c(x) + h^{c(c)}(x), \\ \tilde{h}_{\alpha\beta\gamma\delta}(x) &\equiv h(x) \tilde{\Pi}_{\alpha\beta\gamma\delta}; \quad \tilde{h}_{\alpha\beta\gamma\delta}^c(x) \equiv h^c(x) \tilde{\Pi}_{\alpha\beta\gamma\delta}; \\ \tilde{h}_{\alpha\beta\gamma\delta}^{c(c)}(x) &\equiv h^{c(c)}(x) \tilde{\Pi}_{\alpha\beta\gamma\delta}. \end{aligned} \quad (17)$$

In the following discussion the case of  $I(|x|) = \frac{1}{2} e^{-|x|}$  is considered. An applicability of such an assumption was discussed briefly in [1]. Thus,

$$h^c(x) = \int_0^{+\infty} \frac{1}{2} e^{-|x-x'|} a(x') dx', \quad (18)$$

where  $a(x)$  is the scalar amplitude of tensor  $\langle \tilde{K}_{\alpha\beta\gamma\delta}(x) \rangle$ , i.e.  $\langle \tilde{K}_{\alpha\beta\gamma\delta}(x) \rangle \equiv a(x) \tilde{\Pi}_{\alpha\beta\gamma\delta}$ .



The field  $h^{(e)}(x)$  acts only in a thin surface layer. In order to facilitate the calculations we take the following form for  $h^{(e)}(x)$ :

$$h^{(e)}(x) = \begin{cases} 0 & \text{for } x > 0 \\ c & \text{for } x \leq 0; \quad c - \text{constant.} \end{cases} \quad (19)$$

Taking the second derivative of (17) with respect to coordinate  $x$ , and using (18) and (19) one obtains the following differential equation:

$$h''(x) = \begin{cases} h(x) - a[h(x)], & x > 0 \\ h(x) - C, & x < 0. \end{cases} \quad (20)$$

Let us consider the case when  $x < 0$ . Then the solution of Eqs. (20) is:

$$h(x) = C + D \cdot e^x, \quad (21)$$

and

$$h'(x) = D \cdot e^x = h(x) - C, \quad (22)$$

where

$$D = \int_0^{+\infty} \frac{1}{2} e^{-x} a(x) dx.$$

From (22) one finally obtains the boundary condition on the surface plane:

$$h'(0) = h(0) - C. \quad (23)$$

The applicability of formula (22) at the point  $(x, 0, 0)$  follows from the fact that  $h'(x)$  is at this point a continuous function, (see (18) and (19), where we have taken  $h^{(e)}(0) = 0$ ).

The surface boundary condition, formula (23), was proposed in [1], with  $C$  being a phenomenological constant. Constant  $C$  is connected with the field of elastic deformations near the surface. From formulas (11), (12) and (19) it results that  $C$  is equal to the amplitude of the cubic part of the irreducible tensor  $\varepsilon \cdot \overbrace{u_{\alpha\beta} u_{\gamma\delta}}$ .

The crystal surface is an area where various admixtures assemble as a result of absorption. They also give some contribution to the change of the boundary condition (23). These effects can be treated in an analogous way.

The above considerations show that the method of treating the influence of the surface effects (the elastic deformation, admixtures, etc.) on crystal ordering by the phenomenological constant  $C$ , proposed in [1], can be derived within the general framework of the theory of crystalline order [1, 4, 5].

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